

Action-Angle Clustering of Stellar Streams in Merging Galaxies

MUQING WANG¹ AND TJITSKE STARKENBURG¹

¹*Northwestern University*

ABSTRACT

Stellar streams are sensitive tracers of the Galactic gravitational field and provide a promising probe of the dark-matter distribution in the Milky Way halo. Action-clustering methods exploit the fact that canonical actions are conserved in a time-independent integrable potential. However, the Milky Way is neither static nor isolated, and merger-induced perturbations may alter both the stream population and the reliability of action-clustering as a potential diagnostic.

We test the robustness of radial-action clustering using simulated globular-cluster streams evolved in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations. At each instant, we define a theoretical locally matched spherical reference potential, $\Phi_0(t)$, whose radial force best represents the true acceleration field over the region occupied by a stellar cohort. When $\Phi_0(t)$ is frozen, it assigns every stellar phase-space point a canonical instantaneous reference action, $J_0(t)$. Along the real merger trajectories, these actions evolve through both the residual force not reproduced by $\Phi_0(t)$ and the change of the locally matched reference potential itself.

We identify locally coherent stream cohorts using observable phase-space criteria designed to favor similar instantaneous-action evolution among their members, including smooth along-stream velocity profiles, unfolded radial phase-space structure, limited spatial extent, and locations away from radial turning points. For each selected cohort, we fit a spherical NFW potential by maximizing the one-dimensional Kullback–Leibler divergence of the canonical fitted-model radial actions, $J_{r,\text{fit}}$. At the distance-matched fiducial 1:10 merger snapshot, the selected cohorts provide a useful recovery of the inner-halo enclosed-mass profile. We also compare this fitted distribution with the frozen-BFE radial-cycle proxy distribution, $\tilde{J}_{r,\text{BFE}}$, evaluated from the same stellar phase-space points. The latter characterizes radial orbital structure in the frozen non-integrable potential and need not be a conserved canonical action.

Across the time evolution, stronger mergers generally remove usable coherent material earlier and produce larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable. We decompose the total fitted-to-true mass bias into an exactly additive fitted-to-reference contribution and a reference-to-true contribution. For the 1:10 merger at $20 \leq r/\text{kpc} \leq 30$, broader reference radial-action clouds are associated with a larger positive tail in the fitted-minus-reference KLD and a wider distribution of signed fitted-to-reference mass biases. Earlier snapshots preferentially populate the large-KLD-excess and more extreme mass-bias envelopes. This time ordering suggests that an early, still orderly stream structure can remain susceptible to artificial compression by a mismatched trial potential, whereas later merger-induced irregularity is less readily remapped into a highly clustered action cloud. Trial-potential landscapes further reveal elongated regions with near-equivalent KLD scores and extended regions producing similar median radial actions. These degeneracies make the NFW parameters non-unique, but their mass consequence depends on the direction of the KLD-preferred displacement in parameter space: motion along a nearly constant-enclosed-mass direction can leave the locally inferred mass nearly unchanged, whereas motion across enclosed-mass contours produces a mass bias. Finally, disagreement between the fitted canonical radial actions and the frozen-BFE radial-cycle integrations shows related time dependence to the outer-halo fitted-to-true mass bias. Observable cohort selection can therefore improve the robustness of action-clustering fits, but it cannot guarantee either unique potential recovery or an unbiased global mass profile in a strongly perturbed halo.

Keywords: Galaxies (573) —

1. INTRODUCTION

Stellar streams form through the tidal disruption of globular clusters and dwarf galaxies. Because their stars

originate from dynamically cold progenitors and subsequently follow neighboring trajectories in the Galac-

tic gravitational field, the debris forms thin, coherent structures in phase space. Although a stream may extend over a large spatial region, its small internal phase-space dispersion allows it to retain information about the mass profile, shape, and dynamical evolution of the Milky Way halo.

One way to extract this information is through the distribution of stream stars in action space. In a time-independent integrable Hamiltonian, canonical actions label invariant orbital tori and remain constant along each orbit. Stars stripped from a dynamically cold progenitor are therefore expected to occupy neighboring regions of action space when their phase-space coordinates are evaluated in an appropriate integrable potential. This motivates potential-fitting methods that search for the potential model in which stream actions are maximally clustered. [R. E. Sanderson et al. \(2015\)](#) formulated this approach using the Kullback–Leibler divergence (KLD), and subsequent studies have applied and tested it using mock and observed stream samples (e.g., [S. Reino et al. 2021, 2022](#)). More generally, the construction and numerical accuracy of action coordinates depend on the integrability or approximate separability of the adopted potential ([J. L. Sanders & J. Binney 2016](#)).

Action clustering is nevertheless not an identity between maximum compactness and the correct potential. An incorrect trial potential can shift an action distribution without substantially broadening it, and a short stream segment with limited orbital-phase coverage may not distinguish between competing models. Potential recovery therefore depends not only on the intrinsic coldness of the tracer, but also on its coverage in orbital phase, radius, energy, and angular momentum. The orbital-phase dependence found by [S. Reino et al. \(2021\)](#), together with the improved constraints obtained by combining dynamically distinct streams, illustrates the importance of this sampling problem.

A live merger creates a more fundamental theoretical complication. The actual Galactic Hamiltonian is time dependent and generally non-integrable, so a global set of conserved canonical actions for the full system is not guaranteed to exist. To define a local spherical dynamical target, we introduce at each instant a theoretical reference potential, $\Phi_0(t)$, whose radial force best represents the true acceleration field over the region occupied by a stellar cohort. When $\Phi_0(t)$ is frozen, each actual stellar phase-space point can be assigned to an invariant torus of the reference Hamiltonian and therefore given a canonical instantaneous reference action, $J_0(t)$. As the merger evolves, both the stellar phase-space point and the locally matched reference potential change. The resulting evolution of J_0 contains a physical-response contribution from the actual-minus-reference force and a reference-evolution contribution from the sensitivity of the action to the changing $\Phi_0(t)$. A nearly common change among the members can trans-

late the reference-action cloud without strongly broadening it, whereas member-dependent changes diffuse or distort the cloud. This motivates selecting cohorts whose present-day phase-space structure favors similar instantaneous-action evolution before applying the KLD potential fit.

The Milky Way provides an important example of this non-equilibrium problem. Its halo is neither isolated nor exactly spherical, and it is perturbed by massive satellites and recent accretion events. In particular, the Large Magellanic Cloud (LMC) can generate a dynamical-friction wake and a large-scale response in the Milky Way halo ([C. Vera-Ciro & A. Helmi 2013](#); [N. Garavito-Camargo et al. 2019](#)). Its infall also induces reflex motion between the inner Galaxy and the outer halo, which can bias dynamical inferences that assume a stationary Galactocentric frame ([M. S. Petersen & J. Peñarrubia 2020](#)). Observed and simulated stellar streams exhibit track–velocity misalignments, orbital deflections, and other perturbations associated with the LMC and the deformation of the Milky Way halo (e.g., [D. Erkal et al. 2018](#); [N. Shipp et al. 2019](#); [R. A. N. Brooks et al. 2025](#)).

Recent studies have begun to connect these perturbations directly to action-based potential inference. [A. Arora et al. \(2022\)](#) showed that massive mergers can displace or disrupt stream structures in action space and that time-independent potential representations can lose orbital information on relatively short timescales. Using a time-evolving basis-function-expansion model of the Milky Way–LMC system, [R. A. N. Brooks et al. \(2024\)](#) found that radial-action clustering can retain useful local mass information for shorter-period streams without close LMC encounters, whereas longer-period and strongly perturbed streams may fail to recover the spherically averaged mass profile. These results motivate a distinction between the canonical actions calculated in a fitted integrable model and action-like diagnostics constructed from a more realistic, but potentially non-integrable, potential field.

In this work, we study these limitations using simulated globular-cluster streams evolved in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations. Rather than using all available stream material, we select locally coherent cohorts with smooth along-stream velocity profiles, unfolded radial phase-space structure, limited spatial extent, and locations away from radial turning points. These observable criteria are designed to reject material that is clearly folded, phase-mixed, or kinematically irregular. They do not, by themselves, guarantee an unbiased potential fit. Nevertheless, at the distance-matched fiducial snapshot of the 1:10 merger, the selected cohorts retain useful constraints on the inner enclosed-mass profile.

For each selected cohort, we fit a spherical NFW potential by maximizing the one-dimensional KLD of the

canonical fitted-model radial actions, $J_{r,\text{fit}}$. To diagnose whether this optimization recovers the reference spherical potential, we compare the fitted action cloud with the canonical reference action cloud calculated under $\Phi_0(t)$. We consider both the location of the radial-action distribution and its degree of clustering. In particular, we test whether broadening of the reference action cloud allows a trial potential to cluster the actions more strongly than the reference potential. We also examine the snapshot-time ordering of this relation to test whether broad reference clouds remain equally compressible by the trial-potential family as merger-induced disturbance accumulates.

We also evaluate the KLD and median radial action over a grid of trial NFW parameters. These parameter-plane diagnostics test whether different potentials can produce nearly equivalent KLD scores or similar median radial actions. The former constitutes a KLD-score degeneracy, whereas the latter constitutes a median-action degeneracy. These are distinct forms of non-uniqueness: the first concerns the clustering of the action cloud, while the second concerns its characteristic action value. We additionally compare the directions of these extended regions with the enclosed-mass map. A degeneracy or KLD-preferred parameter displacement produces little mass bias when it follows a nearly constant-mass direction, but it can produce a large mass bias when it crosses enclosed-mass contours. The uniqueness of the NFW parameters and the identifiability of the enclosed mass must therefore be assessed separately.

The total enclosed-mass bias is separated into two contributions. The fitted-to-reference component measures the difference between the cohort-weighted mass profiles of the KLD-best-fit and reference spherical potentials. The reference-to-true component measures the difference between the cohort-weighted reference profile and the globally spherically averaged BFE mass profile. Both components are normalized by the same true enclosed mass and therefore sum exactly to the total fitted-to-true bias. This decomposition distinguishes failure of the KLD fit to recover the reference potential from the additional difference between the reference spherical description and the global mass distribution. Because the two contributions are signed, they may either reinforce or partially cancel one another.

As a separate diagnostic of the full potential structure, we freeze the BFE representation of the simulated halo at each snapshot and integrate the selected stars through one radial cycle. The resulting radial-cycle quantity, $\tilde{J}_{r,\text{BFE}}$, characterizes radial orbital excursions in the frozen, generally non-spherical and non-integrable BFE field and need not be a conserved canonical action. We compare its distribution with that of $J_{r,\text{fit}}$ through their star-count-weighted median displacement and their difference in radial-action KLD. Unlike the fitted-to-reference diagnostics, this comparison probes disagreement between the fitted spherical model and the

radial-orbit mapping generated by the frozen full BFE field over an extended radial region.

Our results show that stronger mergers generally remove useful coherent material earlier and produce larger and more rapidly varying mass biases, while weaker mergers preserve useful constraints for longer. For the 1:10 merger at $20 \leq r/\text{kpc} \leq 30$, increasing reference-action spread is accompanied by an extension toward larger positive fitted-minus-reference KLD and by a widening of the signed fitted-to-reference mass-bias distribution. The time coloring further shows that earlier snapshots preferentially occupy the upper KLD-excess envelope and the more extreme parts of the mass-bias envelope. We interpret this as a difference in how readily the observed kinematics can be remapped: an early broad cloud may still retain an orderly structure that a mismatched potential can compress, whereas later member-dependent disturbance is more difficult for a smooth spherical trial potential to undo. This interpretation is suggestive rather than a unique explanation of the time ordering.

The trial-potential landscapes reveal a second limitation. For the selected snapshot-55 cohort, the near-maximum KLD values form an elongated ridge in the (ρ_0, r_s) plane, while an extended set of parameter combinations produces a median radial action close to the reference value. The KLD score and median action therefore do not uniquely determine the reference NFW parameters. However, the enclosed-mass map shows that the mass consequence of this non-uniqueness depends on direction in parameter space. A KLD-preferred displacement that follows a nearly constant-mass direction can give different NFW parameters but a similar enclosed mass at the radius probed by the cohort; a displacement across mass contours instead changes the inferred mass.

The mass decomposition shows that the fitted-to-reference contribution frequently supplies the dominant structure of the total bias in the illustrated 1:10 inner-halo case, while the reference-to-true contribution is generally more restricted but can become locally substantial. In the outer halo, disagreement between the fitted canonical radial actions and the frozen-BFE radial-cycle integrations shows related time dependence to the fitted-to-true mass bias. These relationships are diagnostic rather than universal: their strength depends on merger history, radial range, snapshot, and the orbital sampling of the retained cohorts.

Section 2 develops the canonical-action, reference-potential, and frozen-BFE radial-cycle framework. Section 3 presents the observable cohort-selection criteria. Section 4 describes the simulations, examines the time-dependent mass recovery, decomposes the total mass bias, and diagnoses over-clustering and degeneracy within the KLD fit. Section 5 discusses the dependence on perturber orbital configuration, the limitations

of observable cohort selection, and directions for future work.

2. THEORETICAL FRAMEWORK

2.1. Canonical Actions and Potential-dependent Action Clustering

For a Hamiltonian system with n degrees of freedom, the canonical action variables are well defined for regular motion confined to invariant tori. In particular, Liouville integrability requires n functionally independent integrals of motion F_i that are mutually in involution,

$$\{F_i, F_j\} = 0 \quad \text{for all } i, j. \quad (1)$$

When the corresponding common level set is regular, compact, and connected, it forms an invariant n -torus. The canonical actions are then defined by integrals over the n independent fundamental cycles γ_i of this torus:

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} \mathbf{p} \cdot d\mathbf{q}. \quad (2)$$

In separable canonical coordinates, this expression reduces to

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} p_i dq_i. \quad (3)$$

Here, q_i and p_i are canonically conjugate variables. The actions label the invariant torus and remain constant under the integrable Hamiltonian, while the conjugate angles evolve along it.

A static spherically symmetric potential is integrable. Away from degenerate orbits, the energy H , total angular momentum squared L^2 , and azimuthal angular momentum L_z provide three functionally independent integrals in involution. Consequently, three canonical actions (J_r, J_θ, J_ϕ) can be defined. Throughout the following expressions, momenta and energies are taken per unit stellar mass. The energy is

$$\begin{aligned} E &= \frac{1}{2} \left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2 \theta} \right) + \Phi(r) \\ &= \frac{1}{2} \left(p_r^2 + \frac{L^2}{r^2} \right) + \Phi(r), \end{aligned} \quad (4)$$

where

$$L^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta}. \quad (5)$$

Since the azimuthal angle ϕ is cyclic in a spherical potential, its conjugate momentum is conserved:

$$p_\phi = L_z. \quad (6)$$

So that the azimuthal action is:

$$J_\phi = \frac{1}{2\pi} \oint p_\phi \cdot d\phi = p_\phi = L_z \quad (7)$$

Using the expression of angular momentum shown in Equ.4, the polar action can be expressed as:

$$J_\theta = \frac{1}{2\pi} \oint p_\theta \cdot d\theta = \frac{1}{\pi} \int_{\theta_{min}}^{\theta_{max}} \sqrt{L^2 - \frac{L_z^2}{\sin^2 \theta}} \cdot d\theta \quad (8)$$

Using $\sin(\theta_{min}) = \sin(\theta_{max}) = \frac{|L_z|}{L}$, the integral gives:

$$J_\theta = L - |L_z| \quad (9)$$

Finally, using Equ.4, the radial action can also be acquired:

$$J_r = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} \sqrt{2[E - V(r)] - \frac{L^2}{r^2}} \cdot dr \quad (10)$$

where r_{min} and r_{max} are the radial turning points in the adopted spherical potential. For the fitted static spherical halo models, we calculate these canonical actions using **Agama**. The code determines the allowed radial interval and evaluates the corresponding canonical actions.

Stars stripped from a dynamically cold progenitor begin on nearby orbits and therefore occupy a relatively compact region in action space when evaluated in an appropriate integrable potential. This compactness contains information about the potential. Stars from the same stream generally sample different orbital phases. *An appropriate potential maps these different phase-space points onto nearby invariant tori*, whereas an incorrect trial potential generally assigns phase-dependent action errors, broadening or distorting the inferred action cloud. This is the physical basis of action-clustering potential inference (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

This statement is not an identity: a wrong potential may translate an action cloud without greatly broadening it, and a short stream segment may provide insufficient phase coverage to distinguish trial models. Potential recovery therefore requires both a dynamically cold tracer and sufficient diversity in orbital phase, radius, energy, or angular momentum. Combining multiple streams on different orbits can reduce individual-stream degeneracies and biases (R. E. Sanderson et al. 2015; S. Reino et al. 2021).

2.2. 1D Kullback-Leibler Divergence

We quantify radial-action clustering using the Kullback–Leibler divergence (KLD). For a radial-action probability density $p(J_r)$ and a reference density $q(J_r)$,

$$D_{KL}[p||q] = \int p(J_r) \ln \left[\frac{p(J_r)}{q(J_r)} \right] dJ_r. \quad (11)$$

where $p(J_r)$ is the probability density of the radial action distribution at the points J_r , and $q(J_r)$ is the background distribution.

Also, following the work by R. E. Sanderson et al. (2015), I use the Modified Breiman Density Estimator to calculate the density profile of the action points. According to B. J. Ferdosi et al. (2011), a fine grid will be built across the action space, and the density value at the center of each fine grid $J_{r,k}$ will be evaluated using the following formula:

$$p(J_{r,k}) = \frac{1}{N} \sum_{i=1}^N (\sigma \lambda_i)^{-d} K_E\left(\frac{J_{r,k} - J_{r,i}}{\sigma_r \lambda_i}\right) \quad (12)$$

Here, i is the star index, N stands for the total number of stars, and d stands for the dimensionality of the action space. This estimator uses the Epanechnikov Kernel K_E instead of a Gaussian Kernel:

$$K_E(t) = \begin{cases} \frac{d+2}{2V_d}(1-t^2), & \text{if } t^2 < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

, where V_d is the volume of the unit sphere in d -dimensional space. We can easily see the the kernel contribution decreases with normalized distance from the sample point to the grid center, which is as expected. Following the analysis in B. J. Ferdosi et al. (2011), the window sizes σ_j in each direction in the action space is derived from coordinate percentiles in each dimension in the action space:

$$\sigma_r = \min\left\{\frac{P_{80}(J_{r,i}) - P_{20}(J_{r,i})}{\ln N}\right\} \quad (14)$$

, where $P_n(J_{r,i})$ is the n th percentile of the radial action values. In addition, define a pilot density $p_{\text{pilot}}(J_{r,i})$ to describe the degree of clustering for the data point by using Equ. 12, with $\lambda_i = 1$. And the adaptive bandwidth λ_i of this estimator is designed to be inversely proportional to the $p_{\text{pilot}}(J_{r,i})$:

$$\lambda_i = \left(\frac{p_{\text{pilot}}(J_{r,i})}{g}\right)^{-\alpha} \quad (15)$$

, where g is the geometric mean of the pilot density:

$$g = \exp\left[\frac{1}{N} \sum_{i=1}^N \ln p_{\text{pilot}}(J_{r,i})\right] \quad (16)$$

and $\alpha = \frac{1}{d}$ is the adaptive-bandwidth exponent. This adaptive bandwidth shrinks kernels in dense regions and widens them in sparse regions, thus increasing the density value where the action points cluster tightly.

Also, for those grids with overly small $p(J_r)$ values, a floor value (like 10^{-9}) needs to be assigned to prevent error. Luckily, the function "modified_breiman_density" in the "MBEtree" package can realize the above calculation for $p(J_r)$.

Following the action-clustering framework of R. E. Sanderson et al. (2015) and its applications by S. Reino

et al. (2021, 2022), trial gravitational potentials are compared through the degree of clustering of the actions calculated from the same observed phase-space data. In this work, the reference distribution $q(J_r)$ is uniform over one fixed and sufficiently wide action interval $[J_{r,\min}, J_{r,\max}]$ that is shared by every trial potential,

$$q(J_r) = \frac{1}{J_{r,\max} - J_{r,\min}}. \quad (17)$$

The interval $[J_{r,\min}, J_{r,\max}]$ is kept the same for all trial potentials. In practice, we choose this universal range to be sufficiently wide to contain the radial actions produced over the full set of potentials considered, including deliberately extreme choices of the potential parameters. This ensures that changes in the KLD are not driven by changing the support of the background distribution from one potential to another.

Thus, the KLD measures how strongly the radial-action distribution departs from a uniform distribution over the fixed reference range. Since the reference interval is held fixed for all trial potentials, maximizing D_{KL}^I is equivalent to finding the potential in which the stream is most concentrated in J_r , or equivalently, has the lowest radial-action entropy. A mismatched potential generally broadens the inferred J_r distribution of a stream, thereby reducing the KLD. The KLD test used here is based on the expectation that the most accurate potential produces the sharpest radial-action clustering relative to the same uniform background. The success of this method has been demonstrated in static test potentials and in applications combining multiple stellar streams (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

2.3. Evolution of Instantaneous Actions in a Locally Matched Spherical Reference Potential

A live merger creates a conceptually different problem from the static integrable case. The actual Hamiltonian,

$$H(\mathbf{z}, t) = \frac{1}{2}v^2 + \Phi(\mathbf{x}, t), \quad (18)$$

is time dependent and generally non-integrable. Consequently, the actual stellar trajectories need not lie on invariant three-tori, and a global set of conserved canonical actions for the merger system is not guaranteed to exist.

Nevertheless, at every instant we may introduce a theoretical spherical and integrable reference potential,

$$\Phi_0(r, t) = \Phi_{\text{sph}}[r; \boldsymbol{\lambda}_0(t)], \quad (19)$$

that best represents the local dynamics experienced by a selected stellar cohort. Here, $\boldsymbol{\lambda}_0(t)$ denotes the parameters of the instantaneous reference potential. This theoretical reference potential is distinct from the potential subsequently obtained by KLD maximization. It

defines the local spherical target that the KLD procedure will later attempt to recover.

More precisely, let $C(t)$ denote the cohort at time t , and let

$$\mathbf{g}(\mathbf{x}, t) = -\nabla\Phi(\mathbf{x}, t) \quad (20)$$

be the true instantaneous acceleration field. A locally matched spherical reference may be defined as the spherical potential whose radial force most closely represents the true force over the region occupied by the cohort:

$$\lambda_0(t) \equiv \arg \min_{\lambda} \frac{1}{N_C} \sum_{a \in C} \|\mathbf{g}[\mathbf{x}_a(t), t] - \mathbf{g}_0[r_a(t); \lambda]\|^2. \quad (21)$$

where

$$\mathbf{g}_0(r; \lambda) = -\frac{\partial\Phi_{\text{sph}}(r; \lambda)}{\partial r} \hat{\mathbf{r}}, \quad (22)$$

and N_C total number of stars in that cohort. Because a spherical model cannot reproduce tangential forces, the relative preference among spherical models in Equation 21 is determined primarily by their ability to reproduce the true local radial acceleration.

The actual Hamiltonian can then be decomposed as

$$H(\mathbf{z}, t) = H_0[\mathbf{z}; \lambda_0(t)] + \delta H[\mathbf{z}, t; \lambda_0(t)], \quad (23)$$

where

$$H_0[\mathbf{z}; \lambda_0(t)] = \frac{1}{2}v^2 + \Phi_0[r; \lambda_0(t)] \quad (24)$$

is the locally matched spherical reference Hamiltonian, and

$$\delta H \equiv H - H_0 \quad (25)$$

contains the part of the true dynamics not reproduced by that spherical reference.

The time evolution of the theoretical reference can be written schematically as

$$\begin{aligned} \Phi_0(t) &= \mathcal{P}_{C(t)}[\Phi(t)], \\ \Phi_0(t+dt) &= \mathcal{P}_{C(t+dt)}[\Phi(t+dt)], \end{aligned} \quad (26)$$

where $\mathcal{P}_{C(t)}$ denotes the locally weighted spherical projection defined by the region occupied by the cohort. The reference potential therefore changes because both the true potential and the locally sampled region evolve. In particular, the residual force at time t changes the subsequent stellar motion and hence the region of the force field occupied by the cohort at $t+dt$.

For a cohort member a , its actual phase-space point at time t can be assigned to an invariant torus of the instantaneous reference Hamiltonian. We define its instantaneous reference action as

$$J_{0,i}^{(a)}(t) \equiv J_i[\mathbf{z}_a(t); \lambda_0(t)]. \quad (27)$$

This is a canonical action of the instantaneously frozen $H_0(t)$, but it is not generally conserved along the star's real trajectory.

The definition in Equation 27 produces a time-ordered sequence,

$$\begin{aligned} J_{0,i}^{(a)}(t) &= J_i[\mathbf{z}_a(t); \lambda_0(t)], \\ J_{0,i}^{(a)}(t+dt) &= J_i[\mathbf{z}_a(t+dt); \lambda_0(t+dt)]. \end{aligned} \quad (28)$$

Both arguments change between the two instants: the true force moves the star to a new phase-space point, while the locally appropriate spherical reference evolves from $H_0(t)$ to $H_0(t+dt)$.

Taking the total derivative of Equation 27 along the actual trajectory gives

$$\begin{aligned} \frac{dJ_{0,i}^{(a)}}{dt} &= \left\{ J_{0,i}^{(a)}, H \right\} + \dot{\lambda}_0^\alpha \left. \frac{\partial J_{0,i}^{(a)}}{\partial \lambda_0^\alpha} \right|_{\mathbf{z}_a} \\ &= - \left. \frac{\partial \delta H}{\partial \theta_{0,i}} \right|_{\mathbf{z}_a} + \dot{\lambda}_0^\alpha \left. \frac{\partial J_{0,i}^{(a)}}{\partial \lambda_0^\alpha} \right|_{\mathbf{z}_a}, \end{aligned} \quad (29)$$

where repeated reference-potential parameter indices α are summed. The first term is the physical response of the star to the actual-minus-reference Hamiltonian. The second term describes the sensitivity of the assigned action to the evolution of the reference potential itself.

When the residual Hamiltonian is purely a potential difference,

$$\delta H = \delta\Phi = \Phi(\mathbf{x}, t) - \Phi_0(r, t), \quad (30)$$

we may define the residual force as

$$\delta \mathbf{F} \equiv \mathbf{g} - \mathbf{g}_0 = -\nabla \delta\Phi. \quad (31)$$

Equation 29 then becomes

$$\frac{dJ_{0,i}^{(a)}}{dt} = \delta \mathbf{F}^{(a)} \cdot \left. \frac{\partial \mathbf{x}}{\partial \theta_{0,i}} \right|_{\mathbf{z}_a} + \dot{\lambda}_0^\alpha \left. \frac{\partial J_{0,i}^{(a)}}{\partial \lambda_0^\alpha} \right|_{\mathbf{z}_a}. \quad (32)$$

The first term measures how the part of the true force not reproduced by $\Phi_0(t)$ changes the instantaneous action. The second term measures how the action changes because the spherical reference used to assign it evolves from $\Phi_0(t)$ to $\Phi_0(t+dt)$.

For two cohort members a and b , Equation 32 gives

$$\begin{aligned} \frac{d}{dt} (J_{0,i}^{(a)} - J_{0,i}^{(b)}) &= \Delta_{ab} \left[\delta \mathbf{F} \cdot \left. \frac{\partial \mathbf{x}}{\partial \theta_{0,i}} \right|_{\mathbf{z}} \right] \\ &\quad + \dot{\lambda}_0^\alpha \Delta_{ab} \left[\left. \frac{\partial J_{0,i}}{\partial \lambda_0^\alpha} \right|_{\mathbf{z}} \right], \end{aligned} \quad (33)$$

where

$$\Delta_{ab} X \equiv X^{(a)} - X^{(b)}. \quad (34)$$

If the total right-hand side of Equation 33 remains similar among the cohort members, their pairwise action separations change slowly. The instantaneous-action cloud can then undergo an approximately semi-rigid

translation through the time-ordered sequence of reference potentials,

$$\Phi_0(t_1) \rightarrow \Phi_0(t_2) \rightarrow \cdots \rightarrow \Phi_0(t_0), \quad (35)$$

without undergoing strong internal diffusion. By contrast, member-dependent residual forcing or member-dependent sensitivity to the changing reference potential broadens or distorts the instantaneous-action cloud.

This distinction is essential for action-clustering inference. The width of an action cloud evaluated in a trial potential may contain both evolutionary decoherence accumulated before the observed snapshot and additional differential action shifts introduced by a mismatch between the trial potential and the locally appropriate $\Phi_0(t_0)$. KLD maximization can be expected to recover $\Phi_0(t_0)$ only when the first contribution has been sufficiently suppressed. Section 3 therefore introduces observable criteria designed to favor cohorts whose present-day phase-space structure is consistent with an approximately semi-rigid instantaneous-action evolution.

2.4. Frozen-snapshot Radial-cycle Action-proxy

On the other hand, if the actual local potential structure differs strongly from being integrable, it will act as an additional factor that declines the accuracy of the KLD potential fitting.

Suppose we know the exact potential model, then at snapshot t_* , we can integrate a model orbit from the measured phase-space position, along which the stellar energy remains constant. If $\tau_{p,k}$ and $\tau_{p,k+1}$ are successive pericentric passages of this frozen model orbit, we define

$$\tilde{J}_{r,\text{BFE}}(t_*) \equiv \frac{1}{2\pi} \int_{\tau_{p,k}}^{\tau_{p,k+1}} p_r(\tau) \dot{r}(\tau) d\tau = \frac{1}{2\pi} \int_{\text{radial cycle}} p_r dr. \quad (36)$$

In an integrable spherical potential this reduces to the canonical radial action. In the frozen, non-spherical merger potential, however, one radial oscillation need not be a fundamental cycle of an invariant three-torus. The quantity may depend on phase, cycle number, and radial-cycle definition. We therefore refer to it as a frozen-potential radial-cycle action proxy rather than a true canonical action.

This proxy does not supply the theoretical basis of the KLD fit and is not an observable input to the method. It is used only as a simulation diagnostic of disagreement between the radial-cycle structure generated by the fitted spherical model and that generated by the frozen potential field.

3. SELECTING COHORTS OF STREAM STARS

The theoretical objective of the cohort selection is to retain stream material for which the instantaneous reference-action cloud is likely to have evolved approximately semi-rigidly through the time sequence of locally

matched spherical reference potentials. For two cohort members a and b , this requires the total member-to-member difference

$$\Delta_{ab} \dot{J}_{0,i}(t) = \dot{J}_{0,i}^{(a)}(t) - \dot{J}_{0,i}^{(b)}(t) \quad (37)$$

to remain sufficiently small over the relevant evolutionary interval. Equivalently, the accumulated change

$$\Delta_{ab} J_{0,i}(t_0) = \Delta_{ab} J_{0,i}(t_1) + \int_{t_1}^{t_0} \Delta_{ab} \dot{J}_{0,i}(t) dt \quad (38)$$

should not strongly enlarge the pairwise action separations.

According to Equation 33, differential instantaneous-action evolution may arise through two channels. First, different cohort members may experience different residual forces relative to the locally matched spherical reference potential. Second, the members may respond differently when that reference potential evolves, because their assigned actions have different sensitivities to its changing parameters.

Neither the true force history nor the full sequence $\Phi_0(t)$ is available in an observation. We therefore construct the cohort selection using only present-day phase-space information. The criteria are designed to favor stream segments whose current morphology and kinematics are expected to accompany small member-to-member differences in the two contributions to $\dot{J}_0$. They are applied before the KLD potential fit to filter out less-informative observed materials.

First, we require a smooth velocity profile along the stream. If the cohort members experience similar residual forces relative to the evolving local spherical reference, their accumulated velocity responses should also remain similar. For the component parallel to the stream,

$$\delta v_{\parallel,a}(t_1 \rightarrow t_0) = \int_{t_1}^{t_0} \delta F_{\parallel,a}(t) dt + \text{const.} \quad (39)$$

A smooth present-day velocity profile is therefore expected when the member-dependent part of this accumulated response is small.

To implement this criterion, we introduce an along-stream coordinate s and fit a quadratic trend to the along-stream velocity:

$$v_{\text{fit}}(s) = As^2 + Bs + C. \quad (40)$$

We retain only cohorts satisfying

$$\Delta_v \equiv \frac{\sqrt{\langle [v_{\parallel}(s) - v_{\text{fit}}(s)]^2 \rangle}}{\text{std}[v_{\parallel}(s)]} < 0.1. \quad (41)$$

Second, we require an unfolded, single-valued phase-space distribution. Members occupying the same

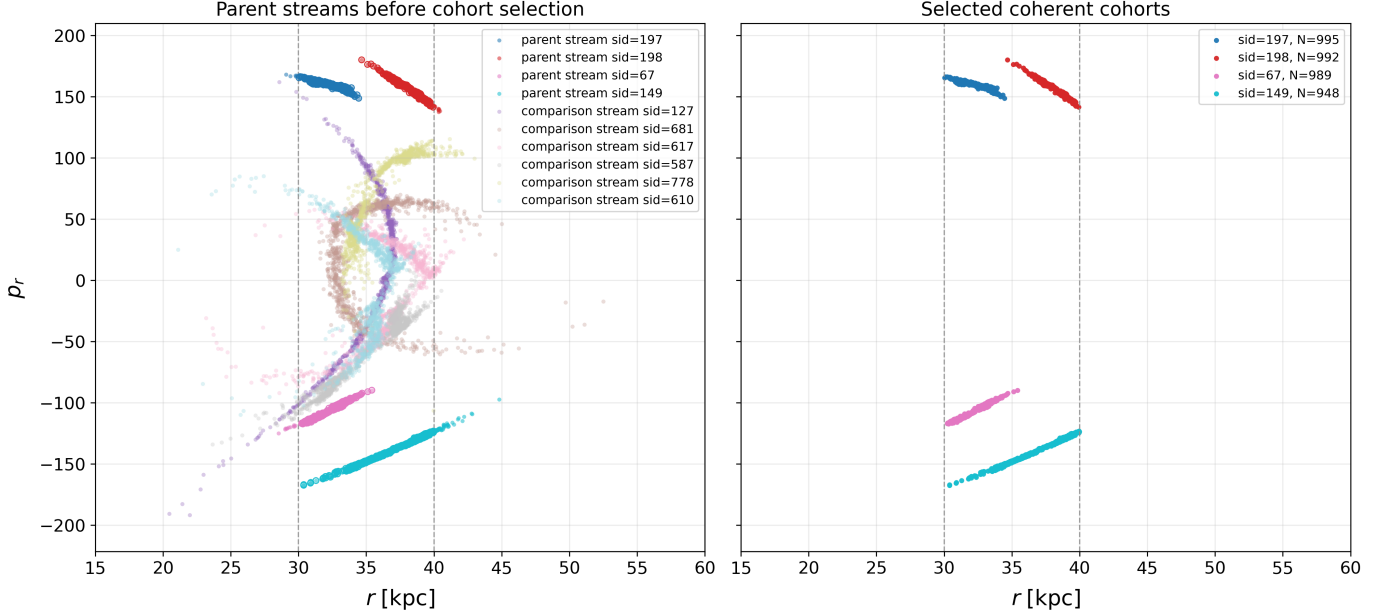


Figure 1. Illustration of the cohort-selection procedure in radial phase space for the 1:10 merger at snapshot 030 in the $30 \leq r/\text{kpc} \leq 40$ radial range. The vertical dashed lines mark the boundaries of the selected radial interval. **Left:** four parent streams together with six comparison streams before cohort selection, including extended, folded, and kinematically irregular phase-space structures. **Right:** the locally coherent segments retained from the four parent streams after requiring a smooth along-stream velocity profile, an unfolded monotonic radial phase-space branch, limited spatial extent, and separation from radial turning points. The listed values of N give the stellar membership of each retained cohort. These criteria favor segments with smaller member-to-member differential forcing and reference-coordinate remapping, but do not by themselves guarantee an unbiased potential fit.

smooth phase-space branch are more likely to have similar reference-orbit tangents,

$$\frac{\partial \mathbf{x}}{\partial \theta_{0,i}}, \quad (42)$$

and similar sensitivities of their instantaneous actions to the evolving reference potential,

$$\left. \frac{\partial J_{0,i}}{\partial \lambda_0^\alpha} \right|_z. \quad (43)$$

These are the quantities multiplying the residual force and the reference-potential evolution in Equation 32. Stars belonging to different wrapped branches may occupy similar spatial locations while having substantially different velocities, radial phases, reference-orbit tangents, and action sensitivities. Their instantaneous actions can therefore respond differently even when they are evaluated under the same sequence $\Phi_0(t)$.

Operationally, we require the fitted velocity trend $v_{\text{fit}}(s)$ to remain strictly monotonic over the selected interval. This removes strongly folded structures in which nearby positions correspond to different orbital branches.

Third, we require the selected cohort to have a limited spatial extent. A short segment occupies a comparatively restricted region of the true force field. It is

therefore more likely that its members experience similar residual forces, $\delta \mathbf{F}$, and that one cohort-common locally matched spherical reference potential provides a meaningful description of their shared radial dynamics. Limited spatial extent also reduces member-to-member variation in both the reference-orbit tangent and the action sensitivity appearing in Equation 32.

At the same time, an extremely narrow radial segment may contain insufficient radial-phase and force-profile information to distinguish among trial spherical potentials. We therefore impose the minimum radial-span requirement

$$\Delta r_{\text{cohort}} \equiv \max_a(r_a) - \min_a(r_a) \geq 2 \text{ kpc}. \quad (44)$$

The upper spatial-extent restriction favors coherent local dynamics, whereas the lower radial-span restriction preserves potential-identification leverage. In the present-day sample analyzed here, all candidate cohorts already exceed the 2 kpc lower limit, so this condition removes no additional present-day cohorts.

Fourth, we exclude cohorts lying close to radial turning points. Near pericentre or apocentre, v_r approaches zero and the mapping between local phase-space coordinates and radial phase becomes poorly conditioned. Stars at similar radii may lie on different incoming and outgoing branches and can consequently have different reference-orbit tangents and different sensitivities to the

evolving reference potential. We therefore impose a threshold on $|v_r|$ to retain cohorts sufficiently far from radial turning points. This requirement is not redundant with the unfoldedness criterion, because a branch may remain single-valued while lying close to a turning point.

The resulting selection criteria are therefore:

1. a smooth along-stream velocity profile satisfying Equation 41;
2. an unfolded phase-space distribution, enforced by the monotonicity of $v_{\text{fit}}(s)$;
3. a limited spatial extent together with the minimum radial span in Equation 44; and
4. sufficient separation from radial turning points, enforced by a threshold on $|v_r|$.

Together, these criteria favor locally coherent segments whose members are expected to experience similar residual-force responses and similar action-coordinate changes as the locally matched spherical reference potential evolves. In the terminology of Section 2.3, they are designed to preserve cohorts for which the instantaneous reference-action cloud can undergo approximately semi-rigid translation rather than strong internal diffusion.

The radial-phase-space comparison illustrates that this selection is deliberately restrictive. Before selection, the parent streams include extended, folded, and kinematically irregular structures, as well as material close to radial turning points. After applying the four criteria, the retained cohorts occupy shorter and more nearly single-valued phase-space branches. These are the cohorts for which the subsequent KLD maximization is more likely to distinguish trial-potential mismatch from action-cloud diffusion already produced by the merger.

The selection does not define the KLD-fitted potential and does not assume that the KLD result is already equal to Φ_0 . Instead, it attempts to establish the dynamical precondition under which maximizing the clustering of trial-model actions can recover the locally appropriate spherical reference potential introduced in Section 2.3.

Fig. 1 illustrates how strongly these criteria reduce the parent-stream population to a small set of locally coherent segments. Before selection, the radial phase-space distribution contains extended streams with substantial curvature, overlap, and folded structures, including material close to orbital turning points. After applying the four selection criteria, the retained cohorts occupy much shorter and more nearly single-valued branches in radial phase space. The selected segments therefore exclude much of the spatially extended, kinematically irregular, and strongly phase-mixed material that would otherwise broaden the inferred action distribution.

This comparison also emphasizes that the selection is deliberately restrictive rather than simply removing a small number of outlying stars. The resulting cohorts represent only those local portions of the streams whose present-day phase-space structure provides strong evidence for coherent evolution history.

4. COMPUTATIONAL WORKS

4.1. Simulation of merging halos with stellar streams

We use the stellar-stream simulations from *Dancing Streams in Merging Halos* (Weerasooriya et al. 2025), a controlled suite designed to study how globular-cluster streams respond to a MW-LMC-like merger. The simulations first generate 1024 mock globular-cluster streams in a static MW-like Hernquist halo using the Fardal et al. (2015) mock-stream distribution function as implemented in Gala. Each stream contains 1000 stellar particles. The stream grid spans two different stellar masses, 10^4 and $10^6 M_\odot$, eight apocenter radii from 30 to 100 kpc, eight circularities from $\eta = 0.5$ to 1.0, and eight stream ages from 0.5 to 6 Gyr, giving 1024 streams in total.

After generation, the same stream population is inserted into live N-body halos and evolved with Gadget4. The host halo has mass $M_{\text{halo}} = 1.57 \times 10^{12} M_\odot$ and Hernquist scale radius $a = 40.85 \text{ kpc}$. The published *Dancing Streams in Merging Halos* suite describes the evolution of the streams in an isolated host and in a 1:5 MW-LMC-like merger, including fiducial, more circular, and radial perturber orbits. In this work, I also use additional related merger data products that extend the stream kinematics to a wider range of perturber-to-host mass ratios, 1:1, 1:2, 1:5, 1:10, and 1:100, with circular and radial perturber-orbit variants available for most ratios. This extended set allows me to study separately the effects of merger strength and perturber orbital geometry on the survival of coherent stream cohorts and on the bias of KLD-based spherical-NFW potential fits.. The full simulations are evolved for 6 Gyr with 241 snapshots. In this work, we analyze snapshots 000–100, corresponding to 0–2.56 Gyr. Our main KLD-fitting analysis focuses on selected cohorts in fixed present-day radial ranges, especially 20–30 kpc, with additional comparisons in 30–40 kpc and 50–70 kpc.

The potential model of the merger halo data packs can be described by the Basis Function Expansion (BFE), which expresses the mass density distribution as a superposition of basis functions:

$$\rho(\mathbf{x}, t) \approx \sum_{n, \ell, m} a_{n, \ell, m}(t) \rho_{n \ell m}(\mathbf{x}). \quad (45)$$

, where $\rho_{n, \ell m}(\vec{x})$ is the basis function that consists of both the radial distribution and the angular distribution:

$$\rho_{n \ell m}(\mathbf{x}) = u_{n \ell}(r) Y_{\ell m}(\theta, \phi). \quad (46)$$

The relationship between the mass density distribution and the potential distribution can be described by Poisson's Equation:

$$\nabla^2 \phi(r) = 4\pi G \rho(r) \quad (47)$$

For those coefficients $a_{n,l,m}(t)$, I adopted the results worked out by David Amir Gonzalez, who is a graduate student at the University of Pennsylvania currently working with Professor Robyn Sanderson, upon their agreement. Their result accurately represent the later-stage potential structure in the simulation.

4.2. Present-day recovery and time-dependent inner-halo performance

4.2.1. Implementing the Selection Criteria

To test those selection criteria, I utilize them to pick coherent cohorts from the tidal streams simulated under an LMC-like merger, which has 1/10 of the mass of the Milky Way halo, merging into the Milky Way. The Milky Way halo is assumed to have a potential described by a spherical NFW model initially. Because the LMC is currently located at a Galactocentric distance of approximately 49.5 kpc (X. Wu et al. 2008) and is likely on its first infall or first passage about the Milky Way (G. Besla et al. 2007; N. Kallivayalil et al. 2013), we identify snapshot 55 ($t = 1.406$ Gyr), at which the simulated host-perturber separation is closest to the observed present-day MW-LMC separation, as the distance-matched present-day analogue used to examine the cohort-selection criteria.

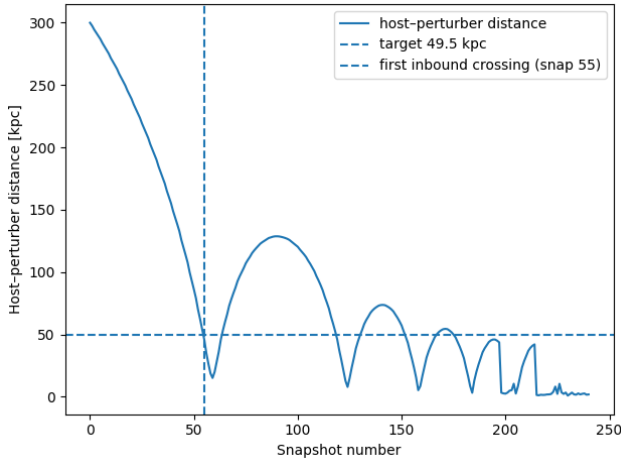


Figure 2. Host-perturber separation as a function of snapshot number for the fiducial Milky Way-LMC-like merger. The horizontal dashed line marks the target separation of 49.5 kpc, and the vertical dashed line identifies its first inbound crossing at snapshot 055 ($t = 1.406$ Gyr). We adopt this snapshot as the distance-matched present-day analogue used for the single-snapshot cohort-selection and enclosed-mass analysis.

In general, the gravitational potential of a perturbed, triaxial MW halo should be described by a general expansion, like the Basis Function Expansion (BFE). However, since I use data from a simulation that assumes a spherical potential initially, and that fitting a general BFE requires a much longer running time along with more stellar members to constrain the parameters, I followed the work of Brooks et al and optimized the parameters of a spherical NFW potential model.

Using those criteria, I selected coherent cohorts, each with at least 100 stars, in every 10-kpc-wide radial range from the galactic center outward. Each coherent cohort will give an optimized NFW model, using which I can integrate for an enclosed mass profile from the galactic center outward. One thing worth noting is that the perturbation force varies so strongly in the region near the center that no coherent cohort can be picked in the region of $r < 10$ kpc.

The spherical enclosed mass implied by an arbitrary gravitational potential can be obtained from the radial gravitational acceleration on a spherical surface. The gravitational acceleration is

$$\mathbf{g}(\mathbf{x}, t) = -\nabla \Phi(\mathbf{x}, t), \quad (48)$$

while Poisson's equation gives

$$\nabla^2 \Phi(\mathbf{x}, t) = 4\pi G \rho(\mathbf{x}, t). \quad (49)$$

Combining Equations 48 and 49, we obtain

$$\nabla \cdot \mathbf{g}(\mathbf{x}, t) = -4\pi G \rho(\mathbf{x}, t). \quad (50)$$

We integrate Equation 50 over the volume V_r enclosed by a sphere of radius r :

$$\begin{aligned} \int_{V_r} \nabla \cdot \mathbf{g} d^3\mathbf{x} &= -4\pi G \int_{V_r} \rho(\mathbf{x}, t) d^3\mathbf{x} \\ &= -4\pi G M_{\text{sph}}(< r, t). \end{aligned} \quad (51)$$

Applying the divergence theorem gives

$$\oint_{S_r} \mathbf{g} \cdot d\mathbf{S} = -4\pi G M_{\text{sph}}(< r, t), \quad (52)$$

where S_r is the spherical surface bounding V_r .

On a sphere of radius r , the outward surface element is

$$d\mathbf{S} = r^2 \hat{\mathbf{r}} d\Omega, \quad (53)$$

and the radial component of the gravitational acceleration is

$$g_r(r, \Omega, t) \equiv \mathbf{g}(r, \Omega, t) \cdot \hat{\mathbf{r}}. \quad (54)$$

Also, we can define the angular average of the radial acceleration at fixed radius as

$$\langle g_r(r, \Omega, t) \rangle_\Omega \equiv \frac{1}{4\pi} \int_{4\pi} g_r(r, \Omega, t) d\Omega. \quad (55)$$

Substituting Equations 53, 54 and 55 into Equation 52 yields the spherically averaged enclosed mass:

$$M_{\text{sph}}(< r, t) = -\frac{r^2}{G} \langle g_r(r, \Omega, t) \rangle_{\Omega}. \quad (56)$$

The above equation does not require the gravitational potential to be spherical. It follows from the total gravitational flux through a spherical surface and therefore returns the mass enclosed by that surface implied by the potential model.

In my numerical calculation, the angular average in Equation 56 is evaluated numerically using $N_{\Omega} = 4096$ nearly uniformly distributed Fibonacci-sphere directions. Thus,

$$\langle g_r(r, \Omega, t) \rangle_{\Omega} \simeq \frac{1}{N_{\Omega}} \sum_{a=1}^{N_{\Omega}} g_r(r, \Omega_a, t), \quad (57)$$

and the enclosed mass is estimated as

$$M_{\text{sph}}(< r, t) \simeq -\frac{r^2}{GN_{\Omega}} \sum_{a=1}^{N_{\Omega}} g_r(r, \Omega_a, t). \quad (58)$$

The Fibonacci construction provides nearly equal-area sampling over the sphere and is sufficiently dense relative to the angular resolution of the NFW spherical model.

In order to improve the precision, I decided to look at the weight-average enclosed-mass profile. In the left panel of Fig.3, I plotted the enclosed mass profile from each cohort only in the radial range where the cohort is located. Then I chopped the whole radial range into 0.1-kpc-wide bins. At each bin r_k , I weight-averaged over the enclosed mass values fitted from each cohort i that covers this bin:

$$M_{\text{fit}}(< r_k, t) = \frac{\sum_{i \in \mathcal{I}_k} N_i M_{\text{sph},i}(< r_k, t)}{\sum_{i \in \mathcal{I}_k} N_i}, \quad (59)$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k , and N_i is the number of selected stars in cohort i . The right panel of Fig. 3 shows the weight-averaged enclosed mass value at each radial bin from the galactic center outward.

From this result, we can see that, despite the error caused by the model simplification, the streams selected by those criteria retrieve the enclosed mass profile reasonably precisely, especially in the range of $r \in [10, 30]$ kpc. Outside this radial range, the discrepancy arises as we go further out, which is expected. At snapshot 55, the center of mass of the merger is about 49.5 kpc away from the host halo. At such a radial distance or outward, a stream segment is more likely to survive as a coherent cohort if it lives in an area farther away from the perturber, where the local Dark Matter particles are more dominant by those from the host halo. Using the fitted

local spherical potential at those lower-density and less-perturbed region to retrieve the spherically-averaged enclosed mass would naturally underestimate the global enclosed mass. So at the current merger stage of the LMC merger, the KLD-maximization method based on those selection criteria discussed above would work well to constrain the potential model inside the radial range sampled by the merger particles.

4.2.2. Time-dependent inner-halo performance

The distance-matched snapshot demonstrates that the selected cohorts can recover a useful enclosed-mass profile at a particular stage of the merger. However, this single-snapshot result does not establish how long the method remains reliable as the halo and stream population evolve. We therefore repeat the cohort selection and spherical-NFW fitting at every snapshot from 0 to 100, corresponding to $0 \leq t \lesssim 2.56$ Gyr.

We first consider cohorts occupying the radial interval $20 \leq r/\text{kpc} \leq 30$. For this inner-halo analysis, the reference enclosed mass is measured directly from the host and perturber dark-matter particles, rather than from the BFE potential.

To characterize the surviving stream sample independently of the tested potential, we define the six-dimensional phase-space compactness of cohort i as

$$C_i = \left[\frac{1}{N_i} \sum_{j=1}^{N_i} \|z_{ij} - \bar{z}_i\|^2 \right]^{1/2}, \quad (60)$$

where j is the star index, N_i is the number of selected stars in cohort i , and

$$z = \left(\frac{x}{10 \text{ kpc}}, \frac{y}{10 \text{ kpc}}, \frac{z}{10 \text{ kpc}}, \frac{v_x}{100 \text{ km s}^{-1}}, \frac{v_y}{100 \text{ km s}^{-1}}, \frac{v_z}{100 \text{ km s}^{-1}} \right). \quad (61)$$

The selected-cohort-weighted compactness is

$$\langle C \rangle_w = \frac{\sum_i N_i C_i}{\sum_i N_i}. \quad (62)$$

At each radial grid point r_k within the 10-kpc-wide radial bin, the weighted fitted enclosed mass is

$$M_{\text{fit}}(< r_k, t) = \frac{\sum_{i \in \mathcal{I}_k} N_i M_{\text{sph},i}(< r_k, t)}{\sum_{i \in \mathcal{I}_k} N_i}, \quad (63)$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k . We quantify the signed enclosed-mass bias by

$$b_M(t) = \left\langle \frac{M_{\text{fit}}(< r_k, t) - M_{\text{true}}(< r_k, t)}{M_{\text{true}}(< r_k, t)} \right\rangle_k \times 100\%. \quad (64)$$

Figure 4 summarizes the resulting evolution calculated under the fiducial 1:10 merger. The first column shows

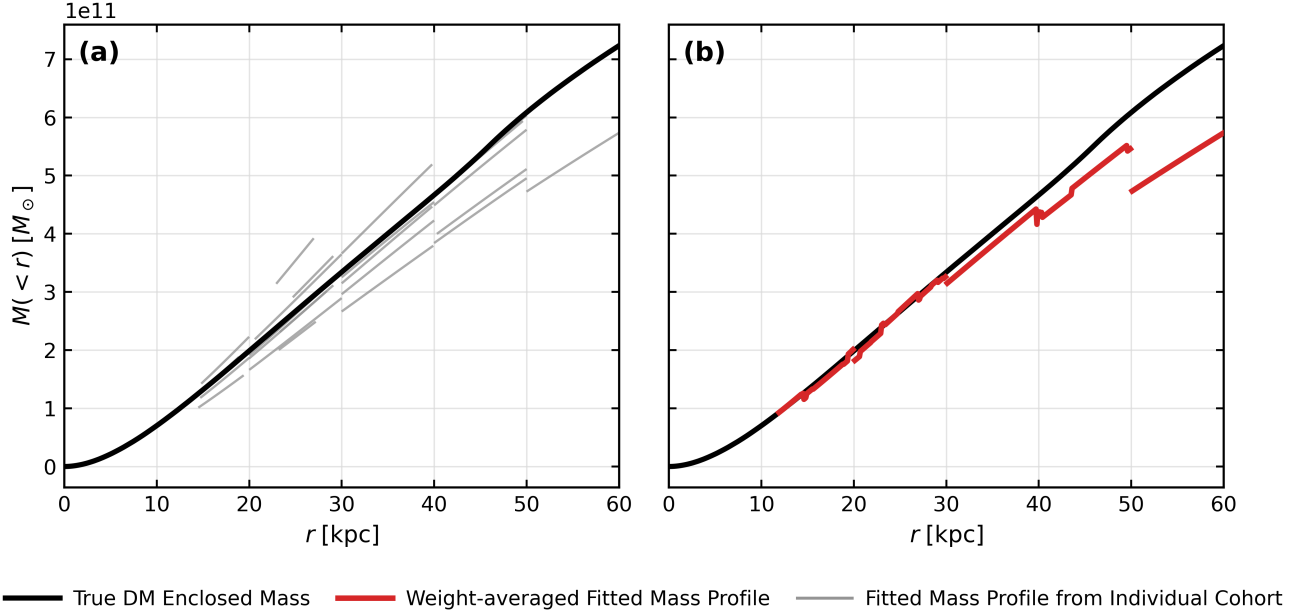


Figure 3. Enclosed-mass recovery from the selected coherent cohorts at snapshot 055, corresponding to the distance-matched present-day configuration of the fiducial 1:10 MW–LMC-like merger. **(a)** KLD-maximized spherical NFW enclosed-mass profiles inferred independently from the individual selected cohorts. Each grey curve is shown only over the radial interval sampled by its corresponding cohort, while the black curve shows the true dark-matter enclosed-mass profile measured directly from the snapshot particles relative to the host center. **(b)** Member-number-weighted mean of the independently fitted cohort mass profiles at each radius (red), compared with the same true enclosed-mass profile (black). Only cohorts whose sampled radial interval includes the given radius contribute to the weighted mean. Thus, the red curve is a radial combination of cohort-specific fits, rather than the result of a single joint fit to all selected stars. The recovered profile agrees most closely with the true profile at smaller radii, while the discrepancy generally increases farther outward. This behavior is consistent with the increasing importance of merger-induced departures from a spherical equilibrium model, together with the decreasing availability and radial coverage of useful coherent cohorts at large radii.

the number of selected cohorts and their mean stellar membership, the second column shows $\langle C \rangle_w$, and the third column shows b_M . The distance-matched present-day reference time is marked by a light-red vertical line.

The number of selected cohorts generally decreases with time in all four merger cases. However, the cohort-count curves do not show a clear monotonic ordering with merger strength. In particular, the stronger mergers do not consistently lose their selected cohorts earlier than the weaker mergers over the full analyzed interval. The compactness generally increases with time, although it does not exhibit a single threshold that separates successful and unsuccessful fits.

The 1:2 merger loses most of its usable cohorts earliest and develops large, rapidly varying mass biases. The 1:5 and 1:10 cases retain selected material for longer, but subsequently exhibit substantial mass fluctuations. The 1:100 merger remains comparatively stable over most of the analyzed interval. Thus, stronger mergers generally shorten the interval over which the fitted mass remains precise.

Occasional late-time snapshots may nevertheless return a small value of b_M . Such isolated agreement does

not by itself demonstrate restored reliability. Because the cohorts are selected independently at each snapshot, the late-time sample can be dominated by an atypical subset of cohorts free of those moderately-perturbed yet passable members. We therefore interpret the time limit of the method from sustained behavior over multiple snapshots, rather than from individual near-zero mass residuals.

These inner-halo results establish the phenomenology of failure, but they do not distinguish why an available cohort produces an incorrect enclosed-mass estimate. We address this question in Section 4.3 using streams in the outer radial range $80 \leq r/\text{kpc} \leq 100$, restricted to snapshot intervals over which the available BFE representation provides a sufficiently reliable description of the simulated force field.

For each selected cohort, we compare the canonical radial-action distribution calculated in the KLD-optimized spherical reference potential with the frozen-BFE radial-cycle action-proxy distribution defined in Equation 36. This comparison diagnoses disagreement between the radial-cycle mappings generated by the optimized spherical model and by the frozen BFE poten-

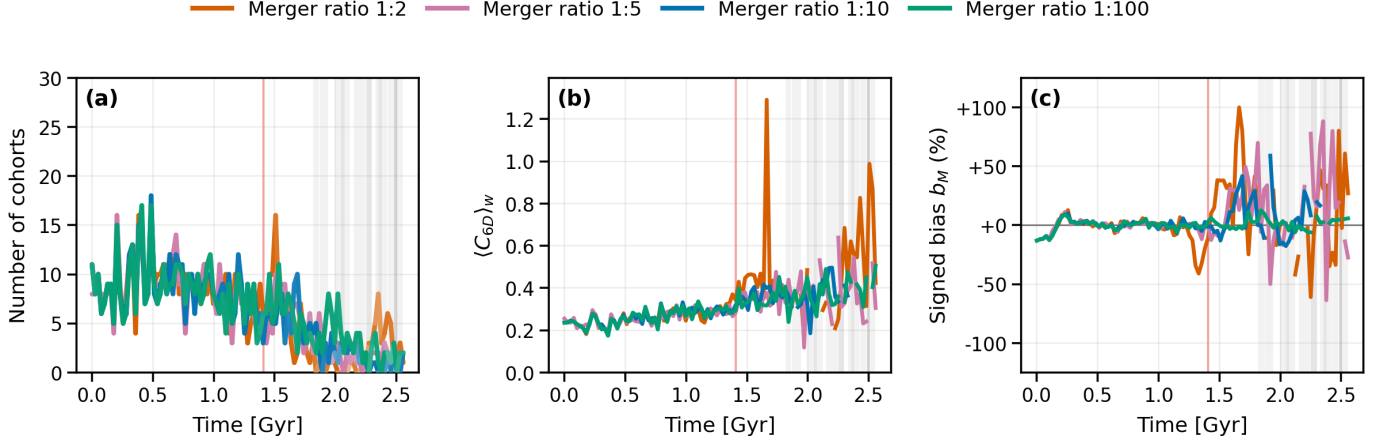


Figure 4. Time-dependent performance of the selected-cohort KLD fitting for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $30 \leq r/\text{kpc} \leq 40$ radial range. The light-red vertical line marks the distance-matched present-day reference time at snapshot 055 ($t = 1.406$ Gyr), defined using the fiducial 1:10 merger. **(a)** Number of selected coherent cohorts. Curve color identifies the merger ratio, while the local color saturation encodes the average stellar membership of the selected cohorts; more saturated segments correspond to larger average membership. **(b)** Cohort-member-number-weighted mean six-dimensional phase-space compactness, $\langle C_{6D} \rangle_w$. **(c)** Signed enclosed-mass bias, b_M , of the cohort-member-number-weighted fitted mass profile relative to the true dark-matter enclosed-mass profile measured directly from the snapshot particles and averaged over the radial grid points in the selected range. The background is white when all four merger cases retain at least one selected cohort and becomes progressively darker grey as fewer cases retain usable cohorts. Stronger mergers generally produce larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable over most of the analyzed interval. The phase-space compactness evolves with time but does not by itself define a universal threshold for successful potential recovery.

tial. We combine this diagnostic with a decomposition of the enclosed-mass bias into fitted-to-local and local-to-global contributions.

4.3. Diagnostics of KLD-fitting reliability

The KLD-fitting procedure assumes that the correct potential can be identified as the one in which the selected stars form the most strongly clustered radial-action distribution. Its reliability therefore depends both on the structure of the reference action cloud and on whether the KLD objective uniquely distinguishes the reference potential from other trial potentials. We identify two mechanisms that can prevent the fit from

recovering the reference potential: over-clustering of an already broadened reference action cloud and degeneracy of either the KLD score or the characteristic action value across the trial-potential grid. These mechanisms are not mutually exclusive and should not be interpreted as additive contributions to the mass error.

The two mechanisms discussed below concern the bias of the fitted potential relative to the reference potential used to calculate the reference actions. The total bias relative to the globally averaged BFE mass profile can additionally contain a reference-to-global contribution. We separate these levels of comparison below before examining the radial-orbit disagreement between the fitted spherical model and the frozen full BFE potential.

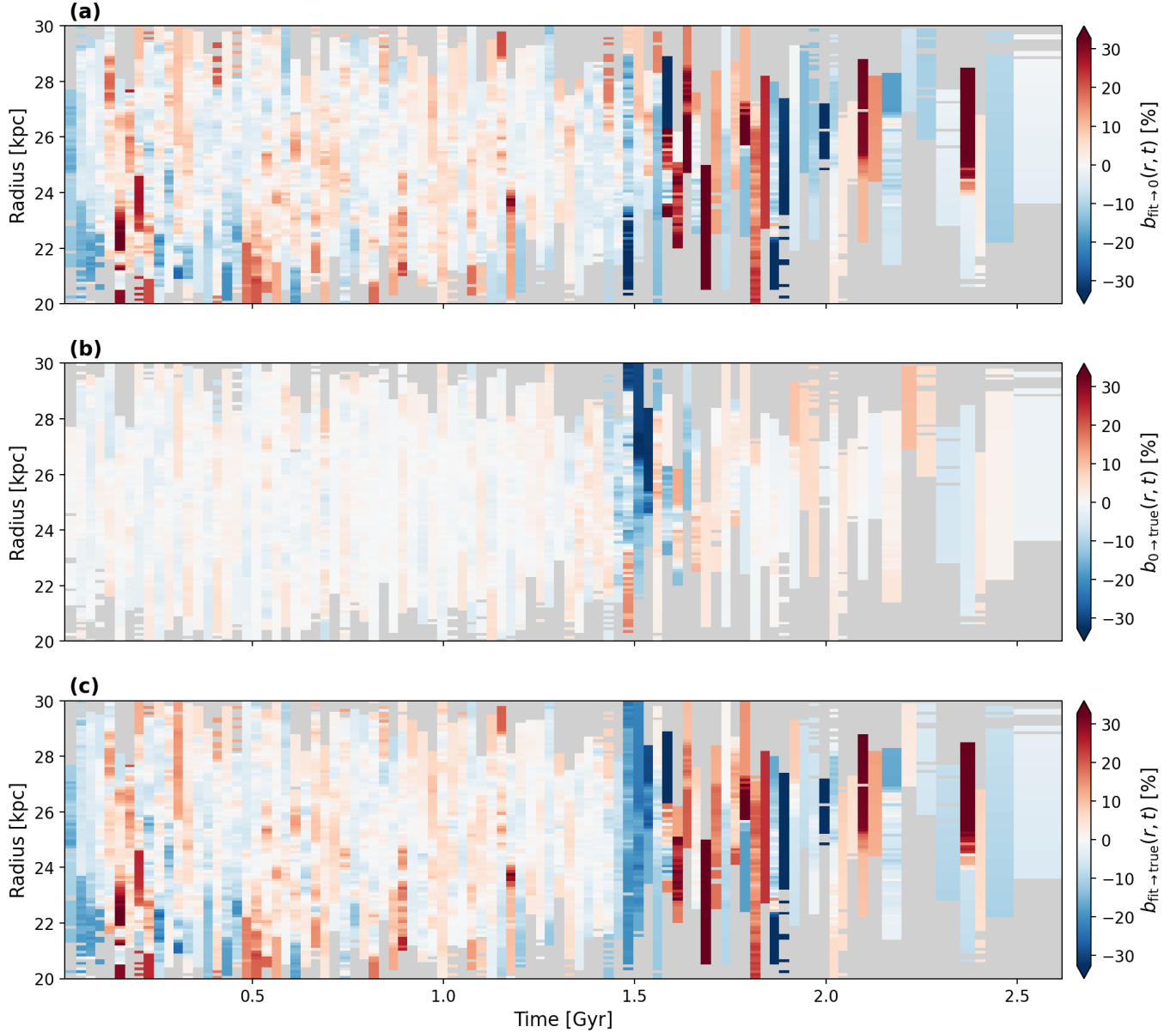


Figure 5. Decomposition of the signed enclosed-mass bias for the 1:10 merger in the $20 \leq r/\text{kpc} \leq 30$ radial range. **(a)** Fitted-to-reference contribution, $b_{\text{fit} \rightarrow 0}(r, t)$, measuring the difference between the cohort-weighted KLD-best-fit and reference spherical mass profiles. **(b)** Reference-to-true contribution, $b_{0 \rightarrow \text{true}}(r, t)$, measuring the difference between the cohort-weighted reference spherical profile and the globally spherically averaged BFE mass profile. **(c)** Total fitted-to-true bias, $b_{\text{fit} \rightarrow \text{true}}(r, t)$. At every valid radius–time pixel, panel (c) is the sum of panels (a) and (b). Positive values indicate an overestimate of the enclosed mass, and negative values indicate an underestimate. Gray regions indicate radius–time pixels for which a valid comparison is unavailable.

4.3.1. Decomposition of the total enclosed-mass bias

Before examining the action-space mechanisms that can make the KLD fit unreliable, we separate the total enclosed-mass bias into two contributions. The first measures whether the KLD-best-fit spherical potential recovers the reference spherical potential. The second measures whether that reference potential reproduces

the globally spherically averaged mass profile of the full BFE field.

Let $M_{\text{fit},w}(< r, t)$ and $M_{0,w}(< r, t)$ denote, respectively, the cohort-weighted enclosed masses of the KLD-best-fit and reference spherical potentials. Let $M_{\text{true}}(< r, t)$ denote the globally spherically averaged enclosed mass measured from the full BFE model. We define the

fitted-to-reference component as

$$b_{\text{fit} \rightarrow 0}(r, t) = 100 \frac{M_{\text{fit},w}(< r, t) - M_{0,w}(< r, t)}{M_{\text{true}}(< r, t)}. \quad (65)$$

This term isolates the mass difference introduced when the KLD fit selects a spherical potential different from the reference potential.

The reference-to-true component is

$$b_{0 \rightarrow \text{true}}(r, t) = 100 \frac{M_{0,w}(< r, t) - M_{\text{true}}(< r, t)}{M_{\text{true}}(< r, t)}. \quad (66)$$

This term remains even if the KLD fit recovers the reference potential exactly. It measures the difference between the reference spherical description sampled by the selected cohorts and the globally averaged mass profile of the full BFE field.

The total fitted-to-true mass bias is

$$b_{\text{fit} \rightarrow \text{true}}(r, t) = 100 \frac{M_{\text{fit},w}(< r, t) - M_{\text{true}}(< r, t)}{M_{\text{true}}(< r, t)}. \quad (67)$$

Because all three quantities use the same $M_{\text{true}}(< r, t)$ denominator, they obey the exact additive decomposition

$$b_{\text{fit} \rightarrow \text{true}} = b_{\text{fit} \rightarrow 0} + b_{0 \rightarrow \text{true}}. \quad (68)$$

No multiplicative cross-term is required under these definitions.

Figure 5 shows the three terms for the 1:10 merger in the $20 \leq r/\text{kpc} \leq 30$ radial range. The fitted-to-reference component in panel (a) contains substantial positive and negative structure across radius and time. The largest and most spatially extended deviations become especially prominent after approximately 1.5 Gyr,

although localized departures are also present at earlier times. These deviations are the part of the total mass error that can be connected directly to failure of the KLD fit to recover the reference spherical potential.

The reference-to-true component in Figure 5(b) is generally smaller and more spatially restricted than the fitted-to-reference component, but it is not negligible. A particularly coherent negative feature appears around $t \simeq 1.5$ Gyr, while additional localized positive and negative differences occur at later times. These structures show that the reference spherical potential does not always reproduce the global BFE enclosed mass even when evaluated over the same selected sample.

The total bias in panel (c) visually resembles the fitted-to-reference component over much of the available radius–time domain. This indicates that failure to recover the reference potential is frequently the larger contribution to the total mass-error structure in this example. Nevertheless, panel (b) modifies the total bias wherever the reference-to-true difference is appreciable. The two signed components can reinforce one another when they have the same sign or partially cancel when they have opposite signs. Consequently, a small total bias at an individual pixel does not necessarily imply that both component errors are small.

To test the numerical closure of the decomposition, we calculate

$$\delta b(r, t) = b_{\text{fit} \rightarrow \text{true}}(r, t) - b_{\text{fit} \rightarrow 0}(r, t) - b_{0 \rightarrow \text{true}}(r, t). \quad (69)$$

Figure 6 shows the resulting radius–time map and the distribution of all valid residuals.

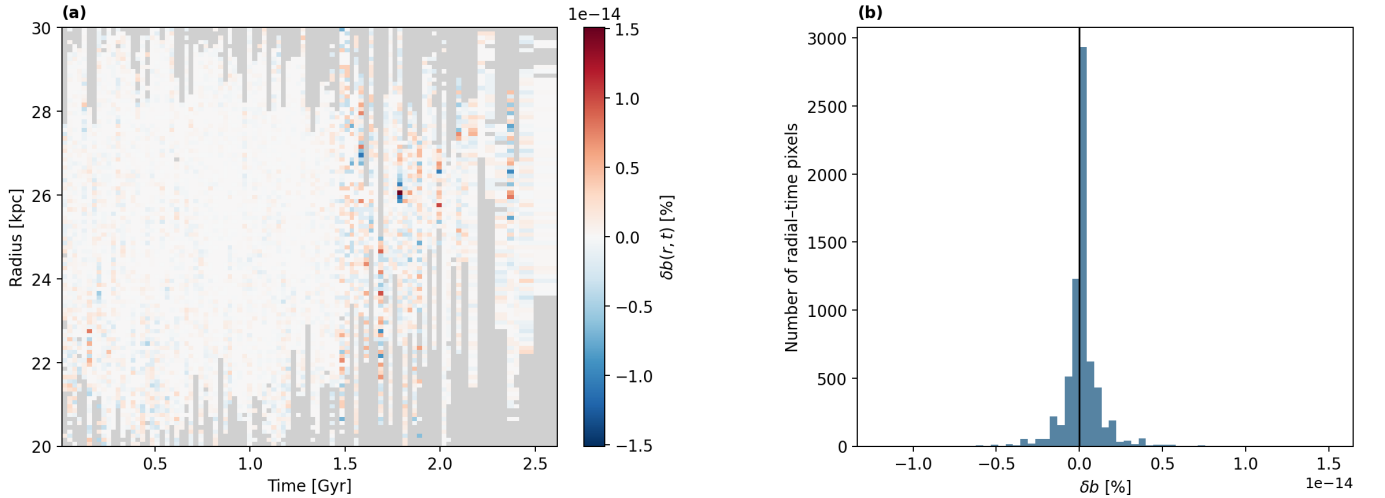


Figure 6. Numerical closure test for the mass-bias decomposition in Figure 5. (a) Closure residual $\delta b(r, t)$ over the valid radius–time pixels. The color scale is of order 10^{-14} percentage points. (b) Distribution of the valid closure residuals. The maximum absolute residual is 1.510×10^{-14} percentage points and the RMS residual is 1.215×10^{-15} percentage points, both far below the adopted 10^{-9} tolerance. Gray regions in panel (a) indicate pixels for which the component biases are unavailable.

The closure residual is consistent with zero to floating-point precision. This confirms that Equation 68 is implemented correctly and that no additional mass-error term is missing from the algebraic decomposition. The closure residual is therefore a numerical validation quantity, not a third physical source of mass bias.

The decomposition also establishes the scope of the following diagnostics. The reference-action spread, KLD over-clustering, KLD-score degeneracy, and median-

action degeneracy concern the origin of $b_{\text{fit} \rightarrow 0}$: they test why the KLD procedure may select a spherical potential different from the reference potential. They do not directly explain $b_{0 \rightarrow \text{true}}$, which instead reflects the difference between the reference spherical description and the globally averaged full BFE mass profile. Moreover, the action-space failure mechanisms are not themselves additive mass-error components and may coexist within the single fitted-to-reference term.

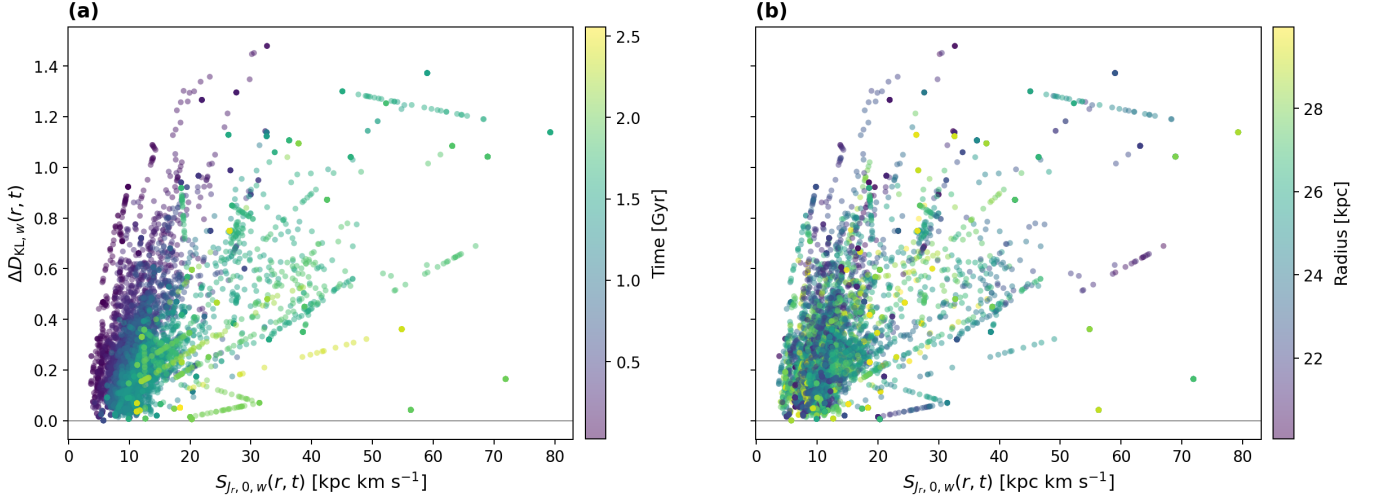


Figure 7. Relation between the weighted reference radial-action spread, $S_{J_{r,0,w}}$, and the weighted KLD excess, $\Delta D_{\text{KL},w}$, for the 1:10 merger in the $20 \leq r/\text{kpc} \leq 30$ radial range. Each point represents one valid radius-snapshot measurement. The points are colored by snapshot time in panel (a) and by radius in panel (b). Positive $\Delta D_{\text{KL},w}$ indicates that the KLD fit produces a more strongly clustered action cloud than the reference potential. The distribution broadens toward larger KLD excess as the reference action cloud becomes more extended, although the relation contains substantial intrinsic scatter. The time coloring in panel (a) shows that earlier snapshots preferentially populate the upper KLD-excess envelope, suggesting that their broad action clouds remain more readily compressible by a mismatched trial potential.

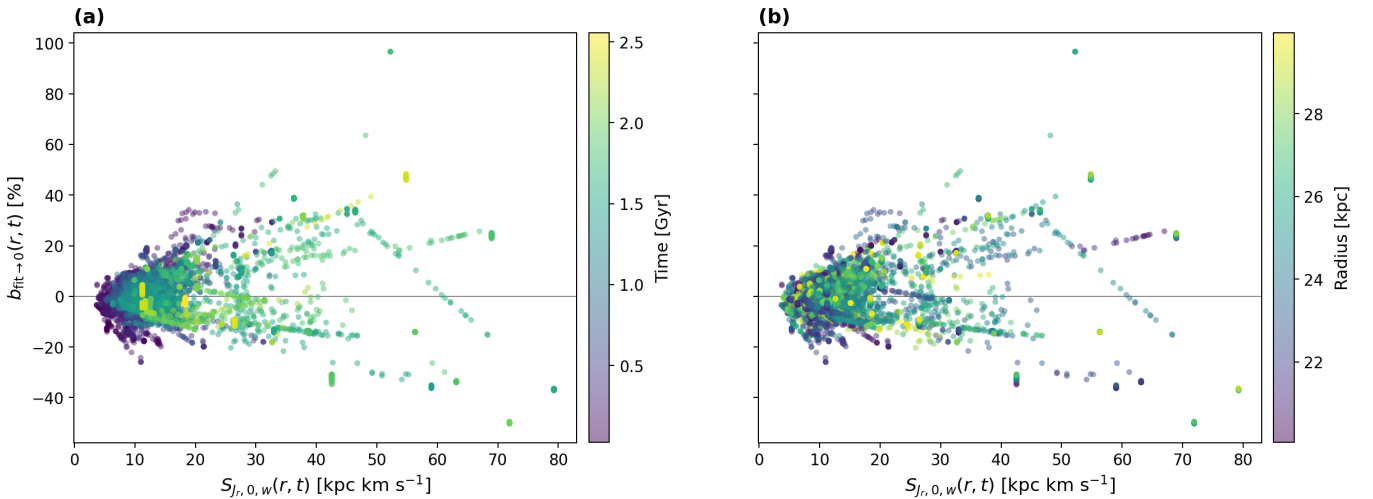


Figure 8. Relation between the weighted reference radial-action spread, $S_{J_{r,0,w}}$, and the signed fitted-to-reference enclosed-mass bias, $b_{\text{fit} \rightarrow 0}$, for the 1:10 merger in the $20 \leq r/\text{kpc} \leq 30$ radial range. The points are colored by snapshot time in panel (a) and by radius in panel (b). The range of permitted signed mass biases widens as the reference action cloud becomes more extended, but the bias can take

either sign. Thus, broadening of the reference action cloud is associated with reduced fitting reliability without uniquely determining the resulting mass error. Earlier snapshots also preferentially occupy the more extreme parts of the mass-bias envelope, although the mass consequence of over-clustering depends on the direction of the fitted displacement in NFW parameter space.

4.3.2. *Over-clustering of a broadened reference action cloud*

The first failure mechanism arises when the selected cohort is not compact in radial action even in the reference potential. We characterize this broadening using the weighted reference-action spread $S_{J_r,0,w}(r,t)$. We then compare it with the weighted KLD excess

$$\Delta D_{\text{KL},w} = D_{\text{KL},w}(\hat{\theta}) - D_{\text{KL},w}(\theta_0), \quad (70)$$

where $\theta * 0$ denotes the reference-potential parameters and $\hat{\theta}$ denotes the KLD-best-fit parameters. Positive $\Delta D_{\text{KL},w}$ means that the fitted potential produces a more strongly clustered action distribution than the reference potential.

The current 1:10 result for $20 \leq r/\text{kpc} \leq 30$ is shown in Figure 7. As $S_{J_r,0,w}$ increases, the occupied region extends toward larger positive $\Delta D_{\text{KL},w}$. The largest KLD excesses occur primarily among the more broadly distributed reference action clouds, although the substantial scatter shows that reference spread alone does not determine how strongly the fitted potential can over-cluster the distribution.

The time coloring in Figure 7(a) adds a further distinction. Earlier snapshots preferentially populate the upper part of the distribution, while later snapshots with broad reference clouds generally show smaller attainable KLD excesses. A simple physical interpretation is that a broad early-time action cloud can still arise from an orderly stream structure. Changing the trial potential can then move the assigned actions of its members coherently and artificially compress the cloud. After further merger evolution, the stellar kinematics contain more member-dependent disturbance. Changing a smooth spherical potential still remaps every assigned action, but it is less able to undo those unequal perturbations and produce a highly clustered distribution. This interpretation concerns how readily the particular phase-space data can be compressed within the adopted NFW family; it does not imply a universal upper bound on KLD at late times.

The corresponding fitted-to-reference mass bias is shown in Figure 8. The distribution develops a broader vertical envelope as $S_{J_r,0,w}$ increases: compact reference clouds are concentrated near small mass bias, whereas broad clouds permit both larger positive and larger negative biases. Earlier snapshots preferentially occupy the more extreme parts of this envelope, consistent with their greater susceptibility to artificial KLD compression. A large KLD excess gives the optimization a stronger preference for a potential away from the reference solution, but it does not by itself determine how much the enclosed mass changes.

Reference-action broadening is therefore a risk indicator rather than a direct predictor of the sign or magnitude of the mass error. Its consequence depends both on how readily the broad cloud can be compressed by the trial-potential family and on the direction in parameter space selected by that compression. The latter dependence is examined in Section 4.3.3.

4.3.3. *KLD-score and action-value degeneracies*

Over-clustering is not the only way for KLD maximization to fail. Even when the fitted and reference action clouds have similar degrees of clustering, the KLD surface need not possess a unique maximum at the reference parameters. We test this possibility by evaluating the radial actions and their KLD over a two-dimensional grid of trial NFW parameters,

$$\theta = (\rho_0, r_s), \quad (71)$$

centered on the reference potential. The same cohort and snapshot are used throughout each parameter grid.

Figure 9 shows the resulting KLD landscape for the selected cohort from parent stream 043 in the 1:10 merger at snapshot 55. The cohort has a median radius of $r_{\text{med}} = 25.17 \text{ kpc}$ and was selected from the $20 \leq r/\text{kpc} \leq 30$ radial range. Rather than forming an isolated peak around the reference potential, the largest KLD values occupy an elongated, approximately one-dimensional region in the (ρ_0, r_s) plane. The near-maximum contour consequently includes multiple combinations of density normalization and scale radius. The KLD score alone cannot strongly distinguish among trial potentials along this direction.

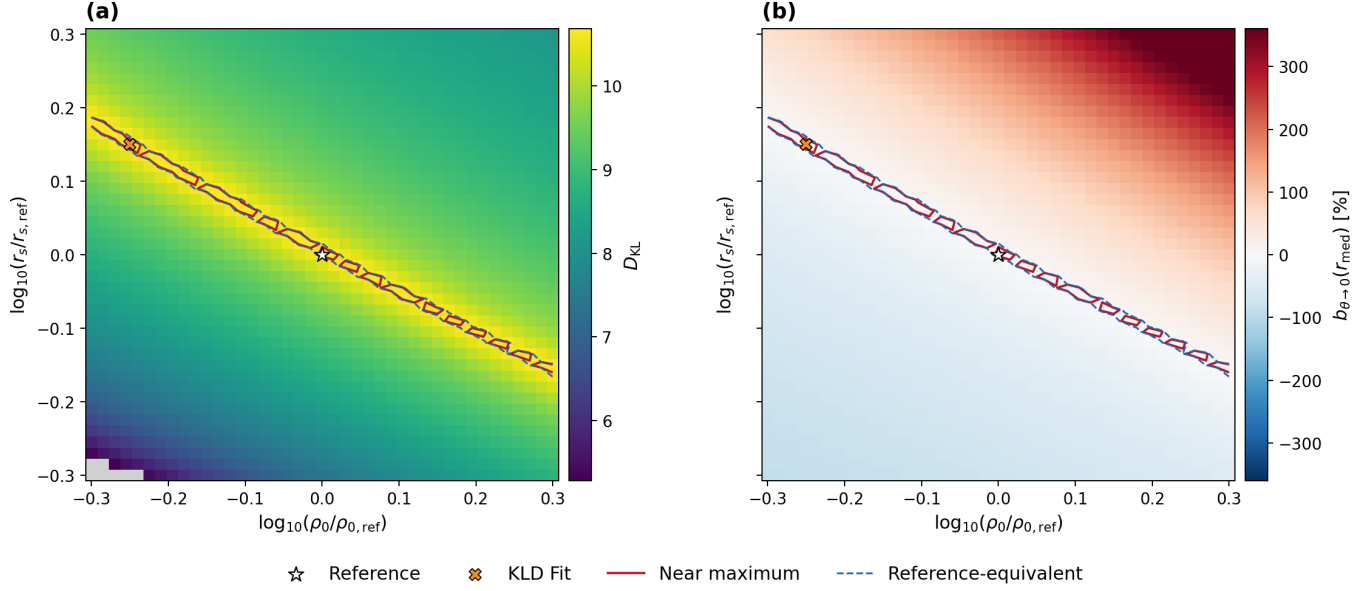


Figure 9. KLD-score degeneracy for a selected cohort from parent stream 043 in the 1:10 merger at snapshot 55. The cohort is selected from $20 \leq r/\text{kpc} \leq 30$ and has median radius $r_{\text{med}} = 25.17 \text{ kpc}$. **(a)** Radial-action KLD evaluated over the trial NFW parameter grid. The axes give the logarithmic offsets of ρ_0 and r_s from their reference values. The solid contour bounds the region whose KLD is indistinguishable from the grid maximum within the adopted finite-sample score tolerance. The dashed contour bounds trial potentials whose KLD is consistent with that of the reference potential under the corresponding matched-star comparison. **(b)** Signed enclosed-mass difference between each trial potential and the reference potential at r_{med} , $b_{\theta \rightarrow 0}(r_{\text{med}}) = 100 \frac{M_{\theta}(<r_{\text{med}}) - M_0(<r_{\text{med}})}{M_{\text{true}}(<r_{\text{med}})}$. The reference potential and the KLD fit are identified by the markers. The extended, nearly one-dimensional KLD contours demonstrate that similar clustering scores can be produced by different NFW parameter combinations. The KLD tolerances define numerical score equivalence and are not confidence intervals. Gray pixels indicate trial potentials for which the finite-action requirement is not satisfied. The mass effect of a parameter displacement depends on whether it follows or crosses the enclosed-mass contours in panel (b).

The companion mass map in panel (b) shows why an extended KLD ridge does not necessarily imply a large enclosed-mass bias. A change in (ρ_0, r_s) changes the inferred mass only to the extent that the parameter displacement crosses the enclosed-mass contours. A displacement along a nearly constant-mass direction can therefore produce substantially different NFW parameters while retaining a similar enclosed mass at the radius sampled by the cohort.

For the illustrated cohort, portions of the near-maximum and reference-equivalent KLD regions follow directions that remain close to $b_{\theta \rightarrow 0}(r_{\text{med}}) = 0$. The KLD score is consequently insufficient to identify unique NFW parameters, but the locally enclosed mass can remain much better constrained than either parameter separately. Conversely, if a KLD-preferred displacement crosses the mass contours, the same parameter non-uniqueness can propagate into a substantial mass bias. The relevant issue is therefore not only the existence of a KLD excess or ridge, but also its orientation relative to the enclosed-mass map.

A second form of non-uniqueness concerns the characteristic action value itself. For every trial potential, we calculate

$$\Delta \text{median } J_r(\theta) = \text{median}_a [J_{r,a}(\theta)] - \text{median}_a [J_{r,a}(\theta_0)], \quad (72)$$

where a indexes the stars used at that parameter-grid location. This quantity measures the displacement of the median radial action relative to the reference potential; it does not measure the width or clustering of the action cloud.

Figure 10 shows that the region producing a reference-like median action is also extended across the NFW parameter plane. Thus, substantially different combinations of ρ_0 and r_s can assign approximately the same median radial action to the selected cohort. The median radial action is therefore not an injective function of the NFW parameters.

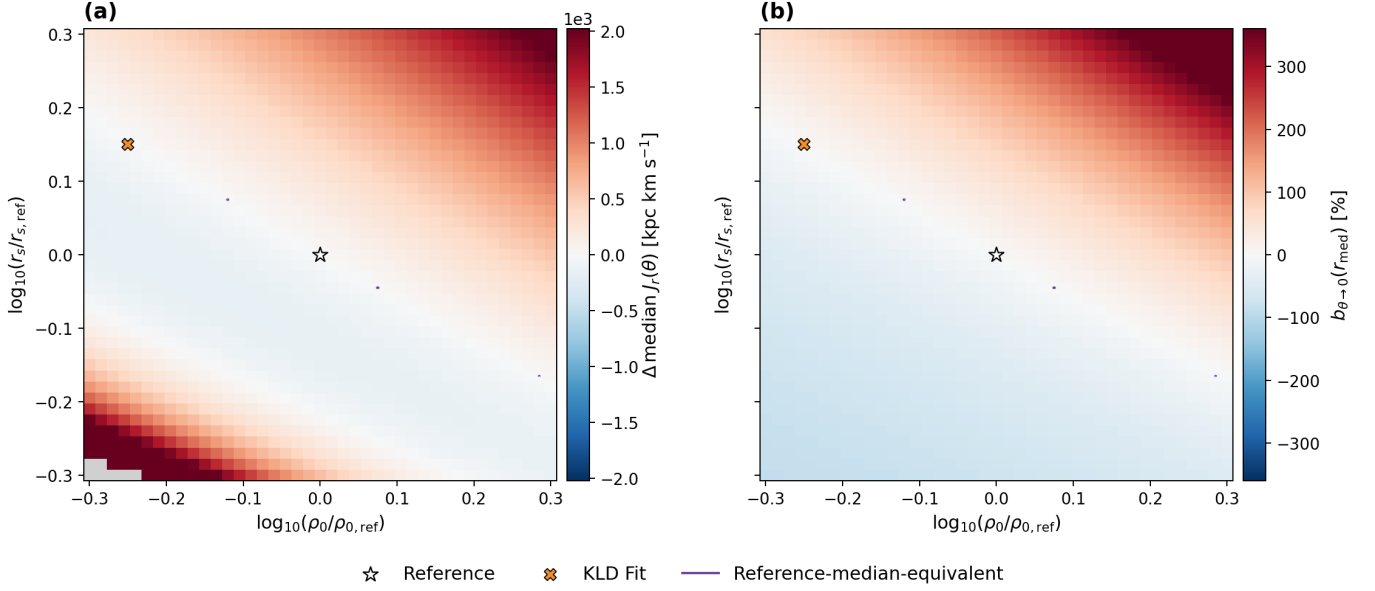


Figure 10. Median-action degeneracy for the same 1:10 cohort, snapshot, and radial range as Figure 9. **(a)** Signed difference between the median radial action in each trial potential and that in the reference potential, $\Delta \text{median } J_r(\theta)$. The contour marks $|\Delta \text{median } J_r| = 0.10 S_{J_r, \text{ref}}$, where $S_{J_r, \text{ref}}$ is the MAD width of the reference radial-action cloud. It therefore identifies the boundary of the region whose median action is close to the reference median under the adopted tolerance. **(b)** Signed enclosed-mass difference $b_{\theta \rightarrow 0}(r_{\text{med}})$ over the same parameter grid. The extended reference-median-equivalent region demonstrates that recovery of the reference median action does not uniquely recover the reference NFW parameters. Gray pixels indicate trial potentials for which the finite-action requirement is not satisfied.

To control numerical truncation by stars with non-finite radial actions, we define a fixed core sample containing stars with finite actions over at least 75% of the trial-potential grid. A KLD value is retained at a given grid point only when at least 95% of this fixed core remains finite there. A separate map of the fraction of stars with non-finite actions is used to verify which parts of the parameter grid are affected by this truncation. This procedure keeps the stellar sample nearly fixed across the accepted grid while avoiding the removal of the entire map because of a small number of locally non-finite orbits.

The two figures diagnose distinct properties of the action cloud. Figure 9 concerns its degree of clustering, whereas Figure 10 concerns the location of its median radial action. A trial potential can therefore obtain a reference-like KLD without reproducing the reference median, or reproduce the reference median without being uniquely identified by the KLD objective. Agree-

ment in both scalar quantities would still not prove equality of the complete action distributions on a star-by-star basis.

These results establish the existence of KLD-score and median-action degeneracies for the illustrated cohort, but they also show that parameter identifiability and mass identifiability are different questions. A KLD-preferred displacement can be large in parameter space yet have little effect on the enclosed mass when it follows a nearly constant-mass direction. Only the component that crosses mass contours changes the inferred mass substantially. The present example demonstrates this directional dependence, but it does not determine how frequently favorable or unfavorable alignments occur across cohorts, snapshots, radial ranges, and merger histories. Over-clustering and degeneracy may also occur simultaneously; they are diagnostic failure routes, not additive components of the mass bias.

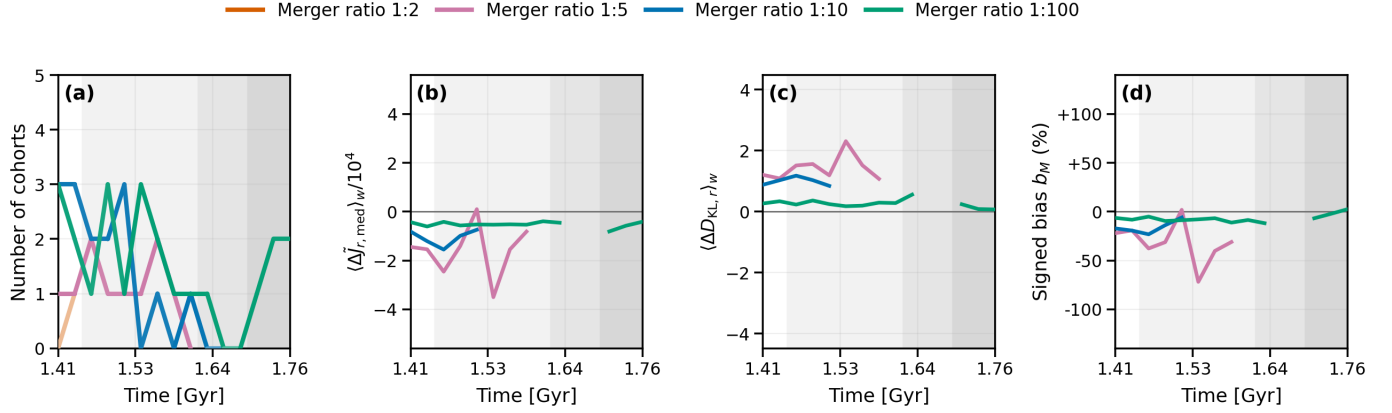


Figure 11. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE field for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $80 \leq r/\text{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\text{Gyr} \leq 1.76$ contribute. (a) Number of selected coherent cohorts. Color identifies the merger ratio, while the saturation of each curve segment encodes the average stellar membership of the selected cohorts. More saturated segments correspond to larger average membership. (b) Matched-star-count-weighted signed median displacement, $\langle \Delta \tilde{J}_{r,med} \rangle_w$, scaled by 10^4 as indicated on the axis. (c) Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{KL,r} \rangle_w$. (d) Signed enclosed-mass bias, b_M , of the fitted spherical model relative to the globally spherically averaged BFE mass profile. The action diagnostics are defined in Equations ?? and ?. White background denotes times at which all four merger cases lie within BFE-valid intervals; progressively darker gray denotes coverage by fewer cases. Missing segments occur when the BFE comparison is unavailable or when no selected cohort contributes a finite measurement.

4.3.4. Radial-orbit disagreement and outer-halo mass recovery

Figure 11 compares the evolution of the selected stream sample, the disagreement between radial actions in the KLD-fitted spherical potential and radial-cycle integrations in the frozen BFE field, and the resulting enclosed-mass bias in the outer-halo range $80 \leq r/\text{kpc} \leq 100$. The action differences and the fitted mass bias exhibit related time dependence. This covariation indicates that disagreement in the radial orbital structure generated by the fitted spherical and frozen full potentials accompanies poor mass recovery. It does not, by itself, establish that the radial-action difference is an independent or additive source of the mass bias. Rather, it provides a dynamical diagnostic of the mismatch already present between the fitted and true orbital mappings.

The figure is restricted to $1.41 \leq t/\text{Gyr} \leq 1.76$, which provides the largest overlapping BFE-valid coverage across the merger cases. This window does not include the principal BFE-valid interval of every individual simulation. In particular, much of the BFE-valid evolution of the 1:2 merger lies outside the displayed interval, so that case contributes only limited information to the comparison. The apparent differences among merger ratios in this restricted window should therefore be interpreted together with their unequal temporal coverage.

5. DISCUSSION AND CONCLUSION

5.1. Dependence on perturber orbital configuration

Figure 12 compares the standard, circular, and radial 1:5 merger configurations over their respective BFE-valid intervals. The radial configuration appears most favorable for the KLD fitting method over the periods analyzed here. Although the close radial encounter may produce a strong transient disturbance, the strongest forcing is plausibly concentrated into a relatively short interval. Outside this interval, coherent cohorts remain or reappear, and the fitted-minus-BFE action discrepancies and enclosed-mass biases are comparatively modest over much of the sampled evolution.

The circular configuration is less favorable in terms of sustained tracer availability. One possible interpretation is that prolonged merger-induced forcing continuously distorts the halo and stream population. The surviving action clouds generally have positive $\Delta D_{KL,r}$ and negative $\Delta \tilde{J}_{r,med}$, indicating that the optimized spherical NFW model produces an action distribution that is more clustered and shifted toward lower radial actions relative to the BFE reference.

This orbital comparison should be regarded as suggestive rather than as a universal ranking of perturber trajectories, since the analysis is restricted to intervals in which BFE provides a sufficiently reliable reference. However, the comparison indicates that not only merger mass ratio, but also the duration and geometry of the encounter, affect the survival of coherent stream material and the form in which the action-space fitting failure appears.

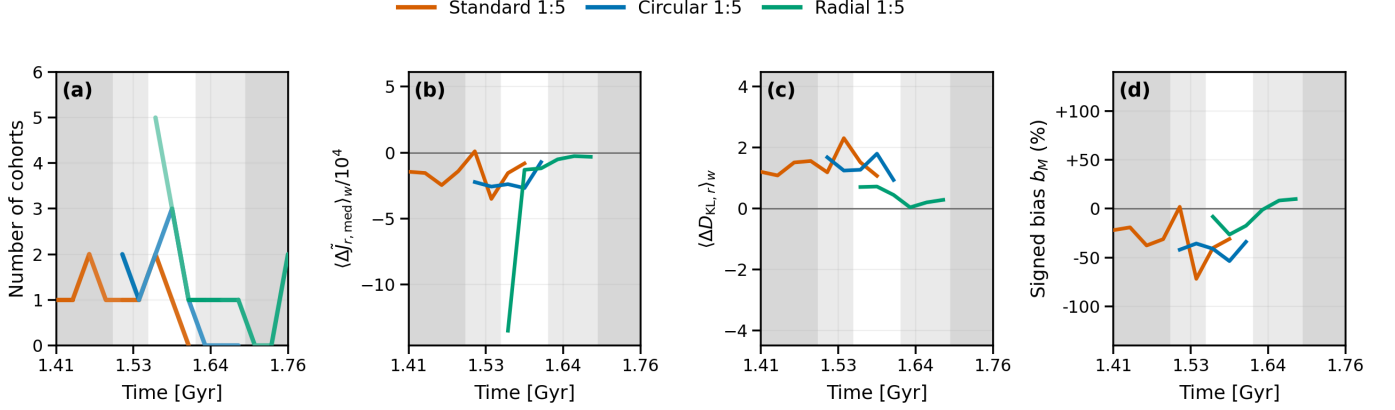


Figure 12. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE reference field for the standard, circular, and radial 1:5 merger configurations in the $80 \leq r/\text{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\text{Gyr} \leq 1.76$ contribute to the comparison. **(a)** Number of selected coherent cohorts. Curve color identifies the orbital configuration, while increasing color saturation represents increasing average stellar membership per cohort. **(b)** Matched-star-count-weighted signed median displacement, $\langle \Delta \tilde{J}_{r,med} \rangle_w$, scaled by 10^4 as indicated on the axis. **(c)** Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{KL,r} \rangle_w$. **(d)** Signed enclosed-mass bias, b_M , of the fitted spherical model relative to the globally spherically averaged BFE reference. White background indicates times at which all three configurations lie within BFE-valid intervals; progressively darker gray indicates that fewer configurations provide a valid BFE comparison. Missing curve segments indicate that no valid diagnostic is available.

5.2. Limitations of observable cohort selection and future prospects

The selection procedure introduced in Section 3 improves the robustness of KLD-based potential fitting by removing stream material that is clearly phase-mixed or kinematically irregular. By retaining short, unfolded stream segments with smooth velocity profiles and locations away from radial turning points, it preferentially selects cohorts whose present-day phase-space structure remains coherent. Nevertheless, coherence is a necessary rather than sufficient condition for reliable potential recovery.

The results of Section 4.3 organize the remaining limitations into different levels. First, a sufficiently perturbed system may cease to contain enough usable coherent cohorts to constrain the potential. Second, even when coherent cohorts remain, the KLD fit may fail to recover the reference spherical potential. A broad reference action cloud can remain sufficiently orderly for another trial potential to compress it artificially, whereas more irregular, member-dependent disturbance can be harder for the adopted spherical family to remove. KLD-score and median-action degeneracies can additionally prevent the reference NFW parameters from being uniquely identified. Whether these effects produce an enclosed-mass bias depends on the direction of the KLD-preferred parameter displacement relative to the enclosed-mass contours. These mechanisms concern the fitted-to-reference mass-bias component and may coexist; they are not independent additive contributions to the mass error.

Third, even exact recovery of the reference potential would not guarantee recovery of the globally averaged true mass profile. The reference-to-true component measures the remaining difference between the reference spherical description sampled by the selected cohorts and the global BFE mass distribution. The mass decomposition shows that this contribution is often more restricted than the fitted-to-reference term in the illustrated 1:10 case, but it can become locally substantial. Because the two contributions are signed, they may reinforce or partially cancel one another. A small total mass bias therefore does not necessarily imply that the KLD fit and the reference-to-true mapping are both individually accurate.

Several diagnostics used here are available only in simulations. The true time-dependent potential, the corresponding canonical reference actions, the direct dark-matter enclosed-mass profile, and the frozen-BFE radial-cycle integrations are not directly observable for the Milky Way. The reference-action spread, fitted-to-reference mass bias, and reference-to-true contribution therefore serve as validation diagnostics rather than quantities that can be measured directly from Galactic stream data.

Other aspects of the reliability analysis can, however, be reformulated as observable or model-based consistency tests. In particular, the shape and stability of the KLD surface over the trial-potential parameters can be evaluated without knowing the true potential. A broad or elongated near-maximum region indicates that the data do not uniquely constrain the adopted potential family, but this parameter degeneracy need not prevent

a useful mass measurement if the region follows a nearly constant-mass direction.

Future applications could therefore supplement the observable cohort-selection criteria with a second-stage, post-fit mass-identifiability screen. For each selected cohort, the set of trial potentials whose KLD scores are indistinguishable from the maximum under a resampling-calibrated score tolerance could be mapped to their enclosed-mass profiles across the radial interval sampled by the cohort. Cohorts for which this near-maximum region spans a narrow enclosed-mass range could be retained or assigned greater weight, whereas cohorts permitting a broad mass range could be downweighted or excluded. This procedure would retain cases in which the NFW parameters are non-unique but the locally enclosed mass remains well constrained, while identifying

degeneracies that propagate directly into mass uncertainty.

Such a post-fit screen would test mass identifiability, not guarantee mass accuracy: the entire near-maximum KLD region could still be displaced from the correct potential through artificial over-clustering. Agreement among independently fitted cohorts, stability under stellar resampling, consistency among different radial ranges or action components, and sensitivity to the adopted fitting bounds would provide complementary checks. Establishing how often KLD-score or median-action degeneracy occurs, how it is oriented relative to enclosed-mass contours, and how well these diagnostics identify unreliable recovery will require a systematic survey across merger histories, radial ranges, snapshots, and cohort properties.

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